

provided support for chemiosmosis as an energy-coupling mechanism.)

3. One of the components of the electron transport chain, ubiquinone (Q), must be able to diffuse within the membrane. It could not do so if the membrane components were locked rigidly into place.

Concept Check 9.5

1. A derivative of pyruvate, such as acetaldehyde during alcohol fermentation, or pyruvate itself during lactic acid fermentation; O_2 ; another electron acceptor at the end of an electron transport chain, such as sulfate (SO_4^{2-}) 2. The cell would need to consume glucose at a rate about 16 times the consumption rate in the aerobic environment (2 ATP are generated by fermentation versus up to 32 ATP by cellular respiration).

Concept Check 9.6

1. The fat is much more reduced; it has many $-CH_2-$ units, and in all these bonds the electrons are equally shared. The electrons present in a carbohydrate molecule are already somewhat oxidized (shared unequally in bonds; there are more C—O and O—H bonds), as quite a few of them are bound to oxygen. Electrons that are equally shared, as in fat, have a higher energy level than electrons that are unequally shared, as in carbohydrates. Thus, fat is a much better fuel than carbohydrate. 2. When you consume more food than necessary for metabolic processes, your body synthesizes fat as a way of storing energy for later use. 3. AMP will accumulate, stimulating phosphofructokinase, and thus increasing the rate of glycolysis. Since oxygen is not present, the cell will convert more pyruvate to lactate, providing a supply of ATP. 4. When O_2 is present, the fatty acid chains containing most of the energy of a fat are oxidized and fed into the citric acid cycle and the electron transport chain. During intense exercise, however, O_2 is scarce in muscle cells, so ATP must be generated by glycolysis alone. A very small part of the fat molecule, the glycerol backbone, can be oxidized via glycolysis, but the amount of energy released by this portion is insignificant compared to that released by the fatty acid chains. (This is why moderate exercise, staying below 70% maximum heart rate, is better for burning fat—because enough O_2 remains available to the muscles.)

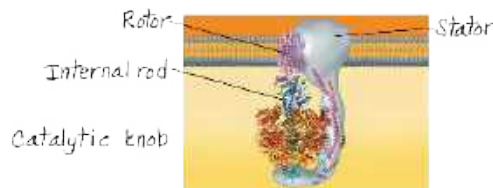
Summary of Key Concepts Questions

9.1 Most of the ATP produced in cellular respiration comes from oxidative phosphorylation, in which the energy released from redox reactions in an electron transport chain is used to produce ATP. In substrate-level phosphorylation, an enzyme directly transfers a phosphate group to ADP from an intermediate substrate. All ATP production in glycolysis occurs by substrate-level phosphorylation; this form of ATP production also occurs at one step in the citric acid cycle. 9.2 The oxidation of the three-carbon sugar, glyceraldehyde 3-phosphate, yields energy. In this oxidation, electrons and H^+ are transferred to NAD^+ , forming NADH, and a phosphate group is attached to the oxidized substrate. ATP is then formed by substrate-level phosphorylation when this phosphate group is transferred to ADP. 9.3 The release of six molecules of CO_2 represents the complete oxidation of glucose. During the processing of two pyruvates to acetyl CoA, the fully oxidized carboxyl groups ($-COO^-$) are given off as 2 CO_2 . The remaining four carbons are released as CO_2 in the citric acid cycle as citrate is oxidized back to oxaloacetate. 9.4 The flow of H^+ through the ATP synthase complex causes the rotor and attached rod to rotate, exposing catalytic sites in the knob portion that produce ATP from ADP and P_i . ATP synthases are found in the inner mitochondrial membrane, the plasma membrane of prokaryotes, and membranes within chloroplasts. 9.5 Anaerobic respiration yields more ATP. The 2 ATP produced by substrate-level phosphorylation in glycolysis represent the total energy yield of fermentation. NADH passes its “high-energy” electrons to pyruvate or a derivative of pyruvate, recycling NAD^+ and allowing glycolysis to continue. In anaerobic respiration, the NADH produced during glycolysis, as well as additional molecules of NADH produced when pyruvate is oxidized, are used to generate ATP molecules. An electron transport chain captures the energy of the electrons in NADH via a series of redox reactions; ultimately, the electrons are transferred to an electronegative atom in a molecule other than oxygen. 9.6 The ATP produced by catabolic pathways is used to drive anabolic pathways. Also, many of the intermediates of glycolysis and the citric acid cycle are used in the biosynthesis of a cell’s molecules.

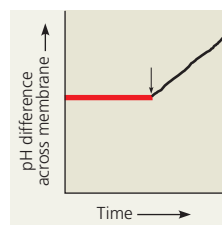
Test Your Understanding

1. C 2. C 3. A 4. B 5. D 6. A 7. B

8. Since the overall process of glycolysis results in net production of ATP, it would make sense for the process to slow down when ATP levels have increased substantially. Thus, we would expect ATP to allosterically inhibit phosphofructokinase. 9. The proton pump in Figures 7.18 and 7.19 is carrying out active transport, using ATP hydrolysis to pump protons against their concentration gradient. Because ATP is required, this is active transport of protons. The ATP synthase in Figure 9.13 is using the flow of protons down their concentration gradient to power ATP synthesis. Because the protons are moving down their concentration gradient, no energy is required, and this is passive transport. 10.



12.



H^+ would continue to be pumped across the membrane into the intermembrane space, increasing the difference between the matrix pH and the intermembrane space pH. H^+ would not be able to flow back through ATP synthase, since the enzyme is inhibited by the poison, so rather than maintaining a constant difference across the membrane, the difference would continue to increase. (Over a longer period of time, the H^+ concentration in the intermembrane space would be so high that no more H^+ would be able to be pumped against the gradient.)

This wasn't asked for in the question, but if your graph levels off at the right end and this is your reasoning, your answer is correct.)

Chapter 10

Figure Questions

Figure 10.11 In the leaf, most of the chlorophyll electrons excited by photon absorption are used to power the reactions of photosynthesis. Figure 10.15 The person at the top of the photosystem I tower would not turn to his left and throw his electron into the NADPH bucket. Instead, he would throw it onto the top of the ramp at his right, next to the photosystem II tower. The electron would then roll down the ramp, get energized by a photon, and return to him. This cycle would continue as long as light was available. (This is why it's called cyclic electron flow.) Figure 10.16 You would (a) decrease the pH outside the mitochondrion (thus increasing the H^+ concentration) and (b) increase the pH in the chloroplast stroma (thus decreasing the H^+ concentration). In both cases, this would generate an H^+ gradient across the membrane that would cause ATP synthase to synthesize ATP. Figure 10.17 Steps that increase $[H^+]$ in the thylakoid space or decrease $[H^+]$ in the stroma contribute to the $[H^+]$ concentration gradient across the thylakoid membrane. In step 2, water is split in the thylakoid space, releasing 2 H^+ and increasing $[H^+]$. In step 3, as electrons travel down the electron transport chain, 4 H^+ are pumped into the thylakoid space, increasing $[H^+]$. In step 5, NADPH formation uses an H^+ from the stroma, decreasing $[H^+]$ in the stroma. Figure 10.22 The gene encoding hexokinase is part of the DNA of a chromosome in the nucleus. There, the gene is transcribed into mRNA, which is transported to the cytoplasm where it is translated on a free ribosome into a polypeptide. The polypeptide folds into a functional protein with secondary and tertiary structure. Once functional, it carries out the first reaction of glycolysis in the cytoplasm.

Concept Check 10.1

1. Because heterotrophs cannot photosynthesize, they cannot capture the energy of sunlight and produce energy-rich compounds like sugars, as autotrophs can. Sugars are oxidized by cellular respiration, providing energy (in the form of ATP) for cellular processes. Without this ability, heterotrophs depend on autotrophs to provide sugars as the food molecules that fuel their vital processes. 2. The main product of fossil fuel combustion (for example, by cars) is CO_2 . Placing containers of algae near sources of CO_2 emission makes sense because the algae need CO_2 to carry out photosynthesis. The higher the CO_2 concentration, the higher will be the rate of algal photosynthesis. (At the same time, algae would be reducing the CO_2 concentration in those areas, which would otherwise contribute to climate change; see Concept 1.1.)

Concept Check 10.2

1. CO_2 enters the leaves via stomata, and, being a nonpolar molecule, can cross the leaf cell membrane and the chloroplast membranes to reach the stroma of the chloroplast. 2. Using ^{18}O , a heavy isotope of oxygen, as a label, researchers were able to confirm van Niel's hypothesis that the oxygen atoms in O_2 produced during photosynthesis comes from H_2O , not from CO_2 . 3. The light reactions could not keep producing NADPH and ATP without the $NADP^+$, ADP, and P_i that the Calvin cycle generates. The two cycles are interdependent.

Concept Check 10.3

1. Green, because green light is mostly transmitted and reflected—not absorbed—by photosynthetic pigments. 2. Water (H_2O) is the initial electron donor; $NADP^+$ accepts electrons at the end of the electron transport chain, becoming reduced to NADPH. 3. In this experiment, the rate of ATP synthesis would slow and eventually stop. Because the added compound would not allow a proton gradient to build up across the membrane, ATP synthase could not catalyze ATP production.

Concept Check 10.4

1. 6, 18, 12 2. The more potential energy and reducing power a molecule stores, the more energy and reducing power are required for the formation of that molecule. Glucose is a valuable energy source because it is highly reduced (has lots of C—H bonds), storing lots of potential energy in its electrons. To reduce CO_2 to glucose, a large amount of energy and a lot of reducing power are required in the form of large numbers of ATP and NADPH molecules, respectively. 3. Yes, it would inhibit the dark reactions. The light reactions require ADP and $NADP^+$, which would not be formed in sufficient quantities from ATP and NADPH if the Calvin cycle stopped.